Ms. Asha Ashokbhai Gondaliya | A.Y.: 2025-26 | Sem./Year: 3 | Course: 2302CS301 - Fundamental of Electronics | Slot Type: Tutorial (TA) | Division: CE-3A

Sr.	Course Content	Planning Title	Planning Description
1	-	Tutorial-1	1. Practice using a basic calculator. **(A)** 2. Practice simple maths formulas. **(A)** 3. Learn how to find remainders using a calculator. **(A)** 4. Understand and practice number base conversions. **(A)** 5. Practice calculations with floating-point numbers. **(A)** 6. Practice exponent (power) calculations. **(A)**
2	Binary Systems	Tutorial -2	1. Decimal to Binary **(A)** (64)10 = (_____)2 (92.29)10 = (_____)2 (26.62)10 = (_____)2 2. Decimal to Octal **(A)** (64)10 = (_____)8 (92.29)10 = (_____)8 (26.62)10 = (_____)8 3. Decimal to Hexadecimal **(A)** (64)10 = (_____)16 (92.29)10 = (_____)16 (26.62)10 = (_____)16 4. Decimal to Binary **(B)** (553.652)10 = (_____)2 (368.772)10 = (_____)2 (543.001)10 = (_____)2 5. Decimal to Octal **(B)** (553.652)10 = (_____)8 (368.772)10 = (_____)8 (543.001)10 = (_____)8 6. Decimal to Hexadecimal **(B)** (553.652)10 = (_____)16 (368.772)10 = (_____)16 (543.001)10 = (_____)16 7. Given that (4096)10 = (1000)x , Find the value of x. **(C)**
3	Binary Systems	Tutorial -3	1. Binary to Decimal **(A)** (111011)2 = (_____)10

			(1110111.010)2 = (___¬_____)10 (1101011.111)2 = (_____)10 2. Binary to Octal **(A)** (111011)2 = (___¬_____)8 (1110111.010)2 = (___¬_____)8 (1101011.111)2 = (_____)8 3. Binary to Hexadecimal **(A)** (111011)2 = (___¬_____)16 (1110111.010)2 = (___¬_____)16 (1101011.111)2 = (_____)16 4. Binary to Decimal (B) (111011.11011)2 = (_____)10 (1110111.0011)2 = (_____)10 (11101100.000110)2 = (_____)10 5. Binary to Octal (B) (111011.11011)2 = (_____)8 (1110111.0011)2 = (_____)8 (11101100.000110)2 = (_____)8 6. Binary to Hexadecimal **(B)** (111011.11011)2 = (_____)16 (1110111.0011)2 = (_____)16 (11101100.000110)2 = (_____)16 7. Convert the decimal number 250.5 to base 3, base 4, base 7 & base 8 **(C)**
4	Binary Systems	Tutorial -4	1. Octal to Decimal **(A)** (45)8 = (_____)10 (66.34)8 = (_____)10 (51.22)8 = (_____)10 2. Octal to Binary **(A)** (45)8 = (_____)2 (66.34)8 = (_____)2 (51.22)8 = (_____)2 3. Octal to Hexadecimal **(A)** (45)8 = (_____)16 (66.34)8 = (_____)16 (51.22)8 = (_____)16 4. Octal to Decimal **(B)** (553.652)8 = (_____)10 (368.772)8 = (_____)10

			(543.001)8 = (_____)10 5. Octal to Binary **(B)** (553.652)8 = (_____)2 (366.772)8 = (_____)2 (543.001)8 = (_____)2 6. Octal to Hexadecimal **(B)** (553.652)8 = (_____)16 (368.772)8 = (_____)16 (543.001)8 = (_____)16 7. Covert Number (4433)5 to Base 10 and Base 2 **(C)**
5	Binary Systems	Tutorial -5	1. Hexadecimal to Decimal **(A)** (A4)16 = (_____)10 (B6D.34)16 = (_____)10 (AE4.40)16 = (_____)10 2. Hexadecimal to Binary **(A)** (A4)16 = (_____)2 (B6D.34)16 = (_____)2 (AE4.40)16 = (_____)2 3. Hexadecimal to Octal **(A)** (A4)16 = (_____)8 (B6D.34)16 = (_____)8 (AE4.40)16 = (_____)8 4. Hexadecimal to Decimal **(B)** (553.652)16 = (_____)10 (368.772)16 = (_____)10 (543.001)16 = (_____)10 6. Hexadecimal to Binary **(B)** (553.652)16 = (_____)2 (368.772)16 = (_____)2 (543.001)16 = (_____)2 7. Hexadecimal to Octal **(B)** (553.652)16 = (_____)8 (368.772)16 = (_____)8 (543.001)16 = (_____)8 8. Convert the decimal number (2598.675)10 to Base 2, Base 4, Base 8, Base 16. **(C)**
6	Binary Systems	Tutorial -6	1. Write down a table of 0 to 7 decimal numbers into binary form and mention odd parity for all numbers.**(A)** 2. Write down a table of 50 to 55 decimal numbers into binary form and mention even parity for all numbers.**(A)** 3. Decimal to BCD **(A)**

			(3441)10 = ( )BCD (2469)10 = ( )BCD (1895)10 = ( )BCD 4. Find the XS-3 code of 37 **(A)** 5. Perform the following**(A)** 1101111 + 110111 11101111 + 110111 11011 – 1101 1110111 – 11011 6. Perform the following**(C)** 111011.101 + 1111.11 110111.110 + 111.101 1010.010 – 111.111 111011.11 – 11110.01
7	Binary Logic And Boolean Algebra	Tutorial -7	1. Simplify following expression: B'C + AB + BC **(A)** A(A' + C) (A'B + C') **(B)** 2. Prove that following expression: A + A'B + AB' = A + B **(A)** (AB + BC + CA)' = A'B' + B'C' + C'A' **(B)** ((A + B')' + A'B' + (A' + B)')' **(C)**
8	Binary Logic And Boolean Algebra	Tutorial -8	1. Prove that following expression: AB + BC + A'C = AB + A'C **(A)** 2. Show that following expression: AB'C + B + BD' + ABD' + A'C = B + C **(A)** (A + C) (A + D) (B + C) (B + D) = AB + CD **(B)** AB + AB'C + BC' = AC + BC' **(C)** 3. Simplify following expression: F = (ABC)' + (AB)'C + A'BC' + A(BC)' + AB'C **(A)** F(A, B, C) = (A + B) (A + C') + A'B' + A'C' **(C)**
9	Boolean Function Implementation	Tutorial -9	1. Simplify following expression using K-Map: F = ∑m(0, 2, 3, 4, 5, 6) **(A)** F(A, B, C, D) = ∑m(0, 1, 2, 5, 8, 9, 10) **(A)** F(A, B, C, D) = ∏M (2, 8, 9, 10, 11, 12, 14**) (A)** F(P, Q, R, S) = ∏M (0, 1, 3, 4, 5, 6, 7, 13, 15) **(B)** F(A, B, C, D) = ∑m(0, 2, 4, 6, 7, 8, 10, 12, 13, 15) **(B)**

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10	Boolean Function Implementation	Tutorial -10	1. F(A, B, C, D) = ∑m(1, 2, 4, 6, 7, 11, 15) + d(0, 3) **(A)** 2. F(W, X, Y, Z) = ∑m(1, 3, 5, 8, 9, 11, 15) + d(2, 13) **(A)** 3. Reduce the expression in SOP and POS form using K-Map: F(A, B, C, D) = ∑m(1, 5, 6, 12, 13, 14) + d(2, 4) **(B)** 4. Reduce using K-Map: F(A, B, C, D) = ∑m(6, 7, 8, 9) + d(10, 11, 12, 13, 14, 15) **(C)**
11	Boolean Function Implementation	Tutorial -11	1. Y= F(W, X, Y, Z) = ∑m (0, 2, 5, 7, 8, 10, 12, 15**)** **(A)** 2. Y= F(A, B, C, D) = ∑m (0, 1, 2, 4, 6, 7, 8, 9, 11, 12, 15) + d(3, 5, 10, 14) **(B)** 3. Y= F(W, X, Y, Z) = ∏M (0, 1, 2, 6, 7, 8, 11, 12, 15) **(A)**
12	Boolean Function Implementation	Tutorial -12	1. Using A-O-I and NAND gate F = ABD + A'C'D' + A'B + A'CD' + AB'D **(A)** F(A, B, C, D)= ABC + A'BC' + CD + AB' **(B)**
13	Basic Combinational Logic	Tutorial -13	1. Using A-O-I and NOR gate F = (A + B' + D) (A' + C' + D') (A' + B + C) (A + C + D') (A' + C' + D) **(A)** F(A, B, C, D) = (A + B + C) (A' + B + D') (A + C + D) (A' + B') **(B)**
14	Basic Combinational Logic	Tutorial -14	1. Implement 3x8 decoder using 2x4 decoder. **(A)** 2. Implement following Boolean function using 8 : 1 multiplexer: Y (A, B, C, D) = ∑m(15, 7, 4, 6, 8, 9, 12, 14) **(A)** F (A, B, C, D) = ∑m(2, 3, 5, 7, 8, 9, 12, 13, 14, 15) **(B)**
15	Basic Combinational Logic	Tutorial -15	1. Implement following Boolean function using 4 : 1 multiplexer: F (A, B, C) = ∑m(1, 3, 5, 6) **(A)** F (A, B, C) = ∑m(1, 2, 4, 7) **(B)** 2. Implement 4x16 decoder using 3x8 decoder. **(C)**